

SAT UZBEKISTAN

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Module-1

1.

A clothing store buys shirts at a wholesale price of 6,00 dollars each and resells them each at a retail price that is 310% of the wholesale price. At the end of the season, any remaining shirts are marked at a discounted price that is 75% of the retail price. What is the discounted price of each remaining shirt, in dollars?

2.

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 136(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 136 gram sample starts to decay. Which statement is the best interpretation of $s(58) = 34$ in this context?

(A)

Approximately 34 grams of the sample remains 58 years after the sample starts to decay.

(B)

The mass of the sample has decreased by approximately 34 grams 58 years after the sample starts to decay.

(C)

The mass of the sample has decreased by approximately 58 grams 34 years after the sample starts to decay.

(D)

Approximately 58 grams of the sample remains 34 years after the sample starts to decay.

3.

Antonia has \$130 in an account. Each year she expects to have 2.5% more money in the account than she had the previous year. Which of the following models best describes how Antonia expects the money in her account to change over time?

(A)

Decreasing exponential

(B)

Decreasing linear

(C)

Increasing exponential

(D)

Increasing linear

4. In the xy -plane, line k and line l are perpendicular and intersect at the point $(2, 9)$. If line k is defined by the equation $y = mx + b$, where m and b are constants and $m > 1$, which of the following points lies on line l ?

(A) $(3, 9 - \frac{1}{m})$

(B) $(3, 9 + \frac{1}{m})$

(C) $(3, 9 - m)$

(D) $(3, 9 + m)$

5.

- In one week in 2017, a technician earned a total of \$900 by working at her regular job and at a second job doing part-time work. The equation $21h + 10c = 900$ represents this situation, where h is the number of hours worked at her regular job and c is the number of hours worked at her second job. Which of the following is the best interpretation of 21 in this context?

(A) The amount, in dollars, the technician earned for each hour she worked at her regular job

(B) The amount, in dollars, the technician earned for each hour she worked at her second job

(C) The number of hours the technician worked in one week at her regular job

(D) The number of hours the technician worked in one week at her second job

6.

- An object is launched from a height of 124 feet above the ground. A quadratic function models the height above the ground, in feet, of the object t seconds after it is launched. According to the model, 3 seconds after the object is launched, it reaches a maximum height of 268 feet above the ground. Which equation defines f ?

(A) $f(t) = -16(t + 268)^2 - 3$

(B) $f(t) = -16(t - 268)^2 + 3$

(C) $f(t) = -16(t + 3)^2 - 268$

(D) $f(t) = -16(t - 3)^2 + 268$

7. $\sqrt[7]{p^4} = x^{\frac{5}{4}}$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{2n-1}$, where n is a constant. What is the value of n ?

8.

A painting studio is offering a promotion where the first class is free, the second class is half off the regular price, and the remaining classes are regularly priced. If the regular price of a class is \$26.40, which function f gives the total cost, in dollars, of x classes taken using this promotion, where $x > 2$?

- (A) $f(x) = 26.40(x - 1) + 13.20$
- (B) $f(x) = 26.40(x - 2) + 13.20$
- (C) $f(x) = 26.40(x - 1) + 13.20(x - 2)$
- (D) $f(x) = 26.40(x - 2) + 13.20(x - 1)$

9.

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side lengths of 10 and 12, which inequality represents the possible lengths, x , of the third side of the triangle?

- (A) $x < 22$
- (B) $x > 22$
- (C) $2 < x < 22$
- (D) $x < 2$ or $x > 22$

10.

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 55 square centimeters (cm^2), and the surface area of right rectangular prism Y is 880 cm^2 . The volume of right rectangular prism Y is $1,600 \text{ cm}^3$. What is the sum of the volumes, in cm^3 , of right rectangular prism X and right rectangular prism Y?

11.

$$125x^2 - (25a + 56)x + ab = 0$$

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k(5a + b)$, where k is a constant, what is the value of k ?

12.

A group of students purchase ice cream cones and ice cream sundaes. The cones cost \$1.40 each, and the cost for 19 sundaes is \$64.60. The total cost for the cones and sundaes is \$85.60. Which equation represents this situation, where x is the number of cones purchased and y is the number of sundaes purchased?

☐ (A) $1.40x + 64.60y = 85.60$

☐ (B) $64.60x + 1.40y = 85.60$

☐ (C) $3.40x + 1.40y = 85.60$

☐ (D) $1.40x + 3.40y = 85.60$

13.

Hakim has a bag of 50 tiles, where 45 are labeled with a letter and 5 are blank. If Hakim selects a tile from the bag at random, what is the probability of selecting a blank tile? (Express your answer as a decimal or fraction, not as a percent.)

14.

A company opens an account with an initial balance of \$35,600.00. The account earns interest, and no additional deposits or withdrawals are made. The account balance is given by an exponential function A , where $A(t)$ is the account balance, in dollars, t years after the account is opened. The account balance after 15 years is \$129,672.38. Which equation could define A ?

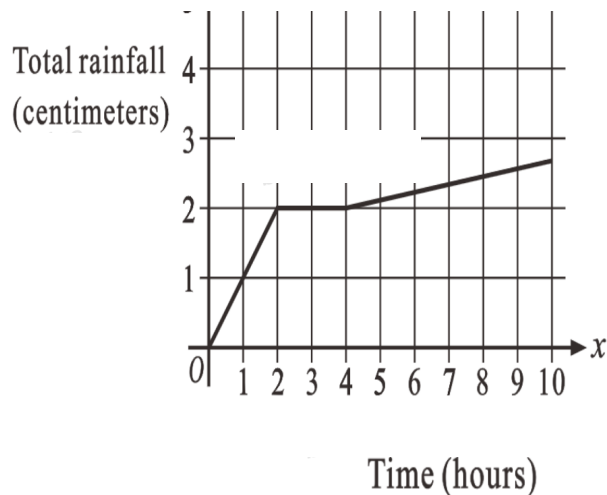
☐ (A) $A(t) = 35,600.00(1.09)^t$

☐ (B) $A(t) = 35,600.00(0.09)^t$

☐ (C) $A(t) = 94,072.38(1.09)^t$

☐ (D) $A(t) = 94,072.38(0.09)^t$

15.



- (A) The rate of rainfall was 2 centimeters per hour between $x = 2$ and $x = 4$.
- (B) The rate of rainfall was the greatest between $x = 4$ and $x = 10$.
- (C) The rate of rainfall increased between $x = 0$ and $x = 2$.
- (D) The rate of rainfall was 0 centimeters per hour between $x = 2$ and $x = 4$.

The graph shows the total amount of rainfall y , in centimeters, from the start of a 10-hour period, where x is the number of hours after the start of the period. Which of the following statements about the rainfall during this time period is true?

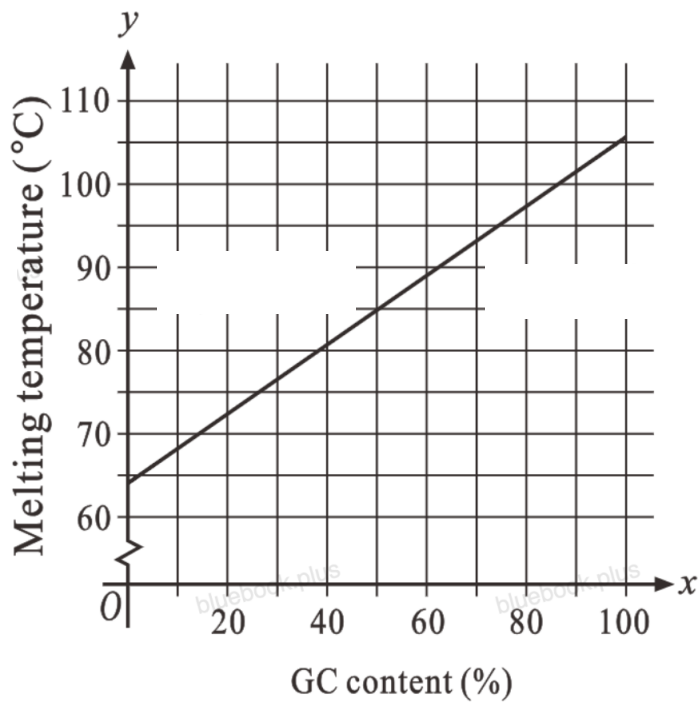
16.

For a set of three consecutive integers, the result of subtracting the greatest integer from the sum of the other two integers is 48. What is the value of the greatest of the three integers?

17.

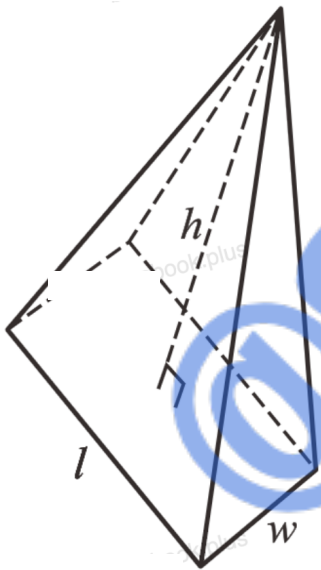
A rectangular banner has an area of 2,300 square inches. A copy of the banner is made in which the length and width of the original banner are each increased by 20%. What is the area of the copy, in square inches?

18.



- (A) For each increase of x by 1, $t(x)$ increases by approximately $\frac{2}{5}$
- (B) A DNA molecule with a GC content of 0% has an estimated melting temperature between 60°C and 65°C .
- (C) A DNA molecule with a GC content of 95% has an estimated melting temperature between 100°C and 105°C .
- (D) A DNA molecule with a GC content of 98% has an estimated melting temperature between 80°C and 90°C .

19.



The figure shown is a right rectangular pyramid, where $l = 18$ units, $w = 9$ units, and $h = 12$ units. What is the surface area, in square units, of the pyramid?

20.

The function $f(x) = \frac{1}{7}(x-6)^2 + 4$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

(A) The toy car's minimum height was 4 inches above the ground.

(B) The toy car's minimum height was 6 inches above the ground.

(C) The toy car's height was 4 inches above the ground when it started moving.

(D) The toy car's height was 6 inches above the ground when it started moving.

21.

A researcher surveyed undergraduate students, graduate students, and postdoctoral students. The number of undergraduate students surveyed was 7,150% of the number of postdoctoral students surveyed and the number of graduate students surveyed was 45% of the number of undergraduate students surveyed. If there were 6,435 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

22.

The bone mineral density of a certain person's thoracic spine was estimated to be 0.812 grams per square centimeter. What is this estimated bone mineral density in grams per square millimeter? (1 centimeter = 10 millimeters)

(A) 81.2

(B) 8.12

(C) 0.0812

(D) 0.00812

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Module-2

1.

A right rectangular prism has a base area of $56t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{14}{3}$ cm, and the height of the rectangular prism is 3 cm.

Which expression represents the surface area, in cm^2 , of the right rectangular prism?

(A) $112t + 56$

(B) $184t + 28$

(C) $1,680t + 28$

(D) $168t$

2.

In the xy -plane, a unit circle with center at the origin O contains point A with coordinates $(1, 0)$ and point B with coordinates

$(\frac{5}{\sqrt{34}}, -\frac{3}{\sqrt{34}})$. If the measure of angle AOB is p radians, what

is the value of $\frac{\cos p}{\sin p}$?

3.

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{334})$. If $h(m) = 0$, what is the greatest possible value of m ?

4.

A computer repair specialist charges \$160 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 6 hours of repair is \$360. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

(A) $f(x) = 50x + 60$

(B) $f(x) = 50x + 160$

(C) $f(x) = 60x$

(D) $f(x) = 60x + 160$

5. A right square pyramid has a height of 12 inches and a volume of 196 cubic inches. What is the perimeter, in inches, of the base of this pyramid?

6.

The value of a share of stock was \$240 on the day it was purchased. Every year, the value of the share of stock increases by 1% of its value the previous year. Which function V gives the value, in dollars, of the share of stock t years after the day it was purchased?

- (A)

$V(t) = 240(1.01t)$
- (B)

$V(t) = 240(1.1t)$
- (C)

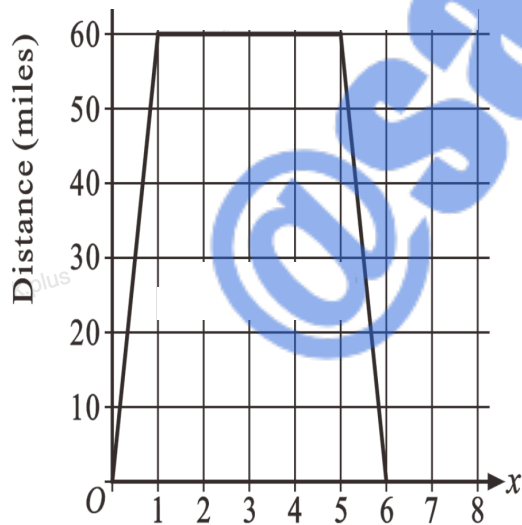
$V(t) = 240(1.01)^t$
- (D)

$V(t) = 240(1.1)^t$

7.

A club plans to sell tote bags. The club members estimate they will sell 80 tote bags when the bags are priced at \$9 each. For every price increase of \$1, they estimate they will sell 8 fewer bags. What is the estimated revenue, in dollars, when the bags are priced at \$12 each? (revenue = price $\times$ number sold)

8.



- (A)

Quinidra drove away from her home at a constant speed for 1 hour, spent 4 hours at the location, then continued to drive away from her home at a constant speed for 1 hour.
- (B)

Quinidra drove away from her home at a constant speed for 1 hour, spent 4 hours at the location, then drove back to her home at a constant speed for 1 hour.
- (C)

Quinidra drove away from her home at an increasing speed for 1 hour, spent 4 hours at the location, then drove back to her home at a decreasing speed for 1 hour.
- (D)

Quinidra drove away from her home at an increasing speed for 1 hour, drove at a constant speed for 4 hours, then drove back to her home at a decreasing speed for 1 hour.

9.

A landscaper will plant two types of shrubs, hydrangeas and winter hazels, in a park. There will be no more than 244 total shrubs planted, and the number of hydrangeas planted will be at most three times the number of winter hazels planted. Which of the following systems of inequalities best represents this situation, where h is the number of hydrangeas that will be planted and w is the number of winter hazels that will be planted?

(A) $h + w \leq 244$ $h \leq 3w$

(B) $h + w \leq 244$ $3h \geq w$

(C) $h + w \geq 244$ $h \leq 3w$

(D) $h + w \geq 244$ $3h \geq w$

10.

$$f(x) = -0.0038400x^2 + 15.236x - 15,103$$

The model shown gives the predicted average Arctic sea ice area $f(x)$, in millions of square kilometers, for September of each year x , where x represents a year and $1979 \leq x \leq 2012$. Based on the model, what is the positive difference, in millions of square kilometers, between the predicted average Arctic sea ice area for September of the year 1991 and the predicted average Arctic sea ice area for September of the year 1992? (Round your answer to the nearest thousandth.)

11.

$$f(x) = x^2 - 4x - 320$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the minimum value of the function as a constant or coefficient?

(A) $f(x) = x^2 - 4(x + 80)$

(B) $f(x) = (x - 2)^2 + (-324)$

(C) $f(x) = x(x - 4) + (-320)$

(D) $f(x) = (x + 16)(x - 20)$

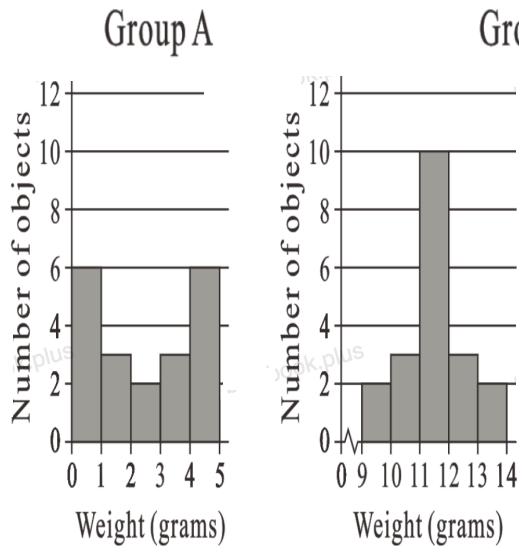
12.

$$g(x) = x^3 + ax^2 + bx + c$$

The function g is defined by the given equation, where a , b , and c are integer constants. The zeros of the function are -2 , -7 , and 6 . What is the value of a ?

13.

The weight of each object in two groups, A and B, was recorded. The histograms represent the distributions of weight, in grams, for the objects in groups A and B.



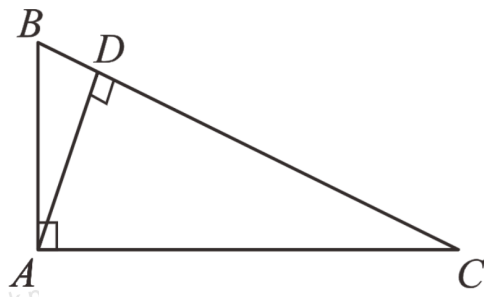
- ☐ A The standard deviation of weight for the objects in group A is less than the standard deviation of weight for the objects in group B.
- ☐ B The standard deviation of weight for the objects in group A is equal to the standard deviation of weight for the objects in group B.
- ☐ C The standard deviation of weight for the objects in group A is greater than the standard deviation of weight for the objects in group B.
- ☐ D There is not enough information to compare the standard deviations of weight for the objects in these two groups.

14.

A carpenter charges a flat rate of \$216 for the first 3 hours of work and \$60 for each additional hour of work. Which equation gives the total amount y , in dollars, that the carpenter charges for x hours of work, where $x > 3$?

- ☐ A $y = 60x + 216$
- ☐ B $y = 216x + 396$
- ☐ C $y = 60x + 36$
- ☐ D $y = 216x + 60$

15.



In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . The measure of $\angle ABD$ is w° , the measure of $\angle CAD$ is z° , and $AD = 178$. If $\cos w^\circ = \frac{9}{41}$, what is the value of $\tan z^\circ$?

16.

A right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. What is the height, in inches, of this pyramid?

17.

$$\frac{1}{cx} = \frac{x}{96} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

18.

A model estimates that a gray whale travels 72 to 77 miles each day during its migration. Based on this model, which inequality represents the estimated total number of miles, x , a gray whale could travel in 27 days of its migration?

☐ (A) $72 + 27 \leq x \leq 77 + 27$

☐ (B) $(72)(27) \leq x \leq (77)(27)$

☐ (C) $72 \leq 27 + x \leq 77$

☐ (D) $72 \leq 27x \leq 77$

19.

At an amusement park, visitors wait in line for a certain ride. People in this line get on the ride at a rate of 14 people per minute. At the end of a certain day, 238 people are in line for the ride when the line closes. If these 238 people will get on the ride and no additional people are allowed to get in line, which function p models the number of people who remain in line t minutes after it closes?

☐ A $p(t) = -14t + 238$

☐ B $p(t) = 14t + 238$

☐ C $p(t) = 14t - 238$

☐ D $p(t) = 252t$

20.

The function f is defined by $f(x) = -29^x$. The function g is a decreasing linear function. In the xy -plane, the graphs of $y = f(x)$ and $y = g(x)$ intersect at two points, (h, j) and (k, m) , where $j > m$. When $g(x) < f(x)$, which of the following must also be true?

☐ A $x > k$

☐ B $x < h$

☐ C $x > k$ or $x < h$

☐ D $h < x < k$

21.

Arturo builds and sells two types of steel tables. The revenue Arturo generates from selling each square table is 2 times his cost to build a square table, and the revenue Arturo generates from selling each hexagonal table is 2.5 times his cost to build a hexagonal table. In one year, Arturo's total cost to build tables was \$25,740, and he generated a total revenue of \$61,875 from selling all the tables he built that year. For that year, how much greater is the revenue Arturo generated from selling hexagonal tables than the revenue he generated from selling square tables?

☐ A \$9,900

☐ B \$15,840

☐ C \$36,135

☐ D \$42,075

22.

$$f(x) = \sqrt{x^3 + ax^2 + bx + c}$$

The function f is defined by the given equation, where a , b , and c are integer constants. In the xy -plane, the graph of $y = f(x)$ passes through the point $(5, 0)$, and $f(0) = 15$. What is the value of $5a + b$?

23.

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(7x - 260)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

☐ (A) $(-14x + 520)^\circ$

☐ (B) $(-14x + 880)^\circ$

☐ (C) $(14x - 520)^\circ$

☐ (D) 180°

24.

The solutions to $x^2 + 6x + 6 = 0$ are r and s , where $r < s$. The solutions to $x^2 + 10x + 13 = 0$ are t and u , where $t < u$. The solutions to $x^2 + 16x + c = 0$, where c is a constant, are $r + u$ and $s + t$. What is the value of c ?

25.

In triangle XYZ , the measure of angle X is 90° . Point W lies on segment YZ , and segment WX is perpendicular to segment YZ . The length of segment YW is 612, and the length of segment WX is 459. What is the value of $\tan Z$?

☐ (A) $\frac{4}{3}$

☐ (B) $\frac{4}{5}$

☐ (C) $\frac{3}{4}$

☐ (D) $\frac{3}{5}$

26.

The mass of object A is 484% of the mass of object B , and the mass of object A is 0.880% of the mass of object C . If the mass of object C is $p\%$ of the mass of object B , what is the value of p ?

27.

For the function f , for each increase in the value of x by c , where c is a positive constant, the value of $f(x)$ increases by a factor of 27. Which of the following equivalent forms of the function f

displays $\frac{1}{c}$ as a coefficient of x ?

(A) $f(x) = 73(3)^{\frac{1}{2}x}$

(B) $f(x) = 73(3^3)^{\frac{1}{6}x}$

(C) $f(x) = 73(9)^{\frac{1}{4}x}$

(D) $f(x) = 73(27^{\frac{1}{3}x})^{\frac{1}{2}}$

28.

The kinetic energy, in joules, of an object with mass 41 kilograms traveling at a speed of v meters per second is given by the function K , where $K(v) = \frac{1}{2}(41)(v^2)$. Which of the following is the best interpretation of $K(0) = 0$ in this context?

(A) The kinetic energy of the object is 0 joules when the mass of the object is 0 kilograms.

(B) The speed of the object is 0 meters per second when the mass of the object is 0 kilograms.

(C) The kinetic energy of the object is 0 joules when the object is traveling at a speed of 0 meters per second.

(D) The height of the object is 0 meters when the object is traveling at a speed of 0 meters per second.

29.

$$x^2 + (\sqrt{k-3})x + 33 = 0$$

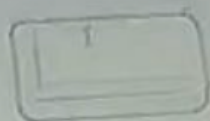
In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k ?

30.

The composition of an animal is defined as the muscles, bones, and fat of the animal. A scientist studied the composition of one young swamp buffalo and determined the buffalo had 123.8 kilograms of muscle, which made up approximately 67.2% of its composition. Of the remaining composition of this buffalo, approximately 47.4% was bone, and the remainder was fat. Based on these approximations, to the nearest tenth, how many kilograms of this buffalo's composition was bone?

31.

The function f is defined by $f(x) = -x^3 - ax^2 - bx - c$, where a , b , and c are integer constants. The zeros of the function are -9 , -4 , and 5 . What is the value of $a + b$?



32.

$$g(x) = x^3 + ax^2 + bx + c$$

The function g is defined by the given equation, where a , b , and c are integer constants. The zeros of the function are -2 , -7 , and 6 . What is the value of a ?

(A) -84

(B) -3

(C) 3

(D) 84

33.

The expression $6x^4 + 23x^2 + 7$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

34.

A square map has a side length of 55 inches, and 1 inch on the map represents an actual distance of 13 miles. A larger version of the same map is printed as a square with the side length 90% longer than the side length of the previous map. On the larger map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

- (A) 1.30
- (B) 6.84
- (C) 24.70
- (D) 49.50

A square map has a side length of 35 inches, and 1 inch on the map represents an actual distance of 17 miles. A smaller version of the same map is printed as a square with the side length 55% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

A) 7.65

B) 10.97

C) 19.25

D) 37.78

35.

$$W(x) = 4x + 80$$

The function W gives the total time, in minutes, it takes for a gardener to water x plants and mow the grass. According to the function, how many minutes does it take the gardener to mow the grass?

36.

A machine makes 8-inch, 9-inch, and 3-inch parts. During a certain day, the number of 8-inch parts that the machine makes is 4 times the number n of 9-inch parts, and the number of 3-inch parts is 30. During this day, the machine makes 100 parts total. Which equation represents this situation?

- (A) $8(4n) + 9n + 3(30) = 100$
- (B) $8n + 9n + 3n = 100$
- (C) $4n + 30 = 100$
- (D) $5n + 30 = 100$

37.

The solutions to $x^2 + 6x + 6 = 0$ are r and s , where $r < s$. The solutions to $x^2 + 10x + 13 = 0$ are t and u , where $t < u$. The solutions to $x^2 + 16x + c = 0$, where c is a constant, are $r + u$ and $s + t$. What is the value of c ?

38.

For the function f , for each increase in the value of x by c , where c is a positive constant, the value of $f(x)$ increases by a factor of 27. Which of the following equivalent forms of the function f displays $\frac{1}{c}$ as a coefficient of x ?

(A) $f(x) = 73(3)^{\frac{1}{2}x}$

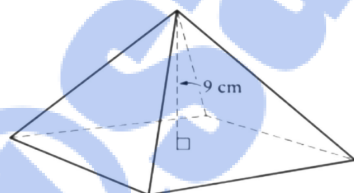
(B) $f(x) = 73(3^3)^{\frac{1}{6}x}$

(C) $f(x) = 73(9)^{\frac{1}{4}x}$

(D) $f(x) = 73(27^{\frac{1}{3}x})^{\frac{1}{2}}$

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39.



Note: Figure not drawn to scale.

The right square pyramid shown has a height of 9 centimeters (cm). The length, in cm, of one edge of the pyramid's base is $\frac{13}{3}$ times the height of the pyramid. What is the volume, in cm^3 , of the pyramid?

40.

Data Set X	
Value	Frequency
a	0
b	5
c	19

Data Set Y	
Value	Frequency
a	1,900
b	500
c	0

Data sets X and Y are summarized in the tables shown, where a , b , and c are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

- (A) The mean of data set X is $\frac{2(19)}{24}$ less than the mean of data set Y.
- (B) The mean of data set X is $\frac{2(19)}{24}$ greater than the mean of data set Y.
- (C) The mean of data set Y is 100 times the mean of data set X.
- (D) The means of the two data sets are equal.

41.

The amount of money owed on a certain type of loan has two components: the principal balance of the loan and the amount of interest accrued on the loan. For a loan of this type, each payment was \$298.00 and there were 60 payments total. Part of each payment was applied to the principal balance of the loan, and the rest was applied to the amount of interest accrued on the loan. The amount of the 1st payment that was applied to the principal balance was \$220.93. For each additional payment, the amount that was applied to the principal balance was approximately 0.5% greater than the amount that was applied to the principal balance for the previous payment. Which function best approximates the amount of the x th payment, in dollars, that was applied to the amount of interest accrued, where $x \leq 60$?

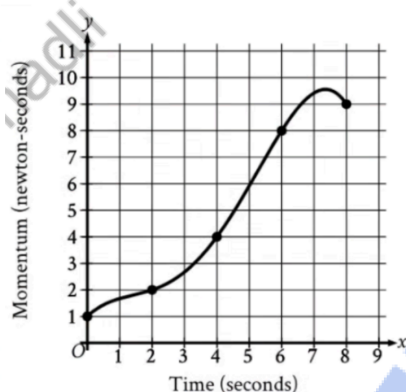
- (A) $f(x) = 298.00 - 220.93(1.005)^{x-1}$
- (B) $f(x) = 298.00 - 220.93(1.005)^x$
- (C) $f(x) = (298.00 - 220.93)(0.995)^{x-1}$
- (D) $f(x) = (298.00 - 220.93)(0.995)^x$

42.

A company developed a plan to set the selling price of a product. The company determined that for a selling price of \$150.00, zero products would be sold. For each \$2.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly \$2,187.50, which of the following could be the number of products sold? (revenue = price $\times$ number of products sold)

- (A) 25
- (B) 30
- (C) 815
- (D) 2,250

43.



The graph shows the momentum y , in newton-seconds, of an object x seconds after the object started moving, for $0 \leq x \leq 8$. What is the average rate of change, in newton-seconds per second, in the momentum of the object from $x = 2$ to $x = 6$?

44.

$$y = 2x^2 - 23x + 64$$

$$y = x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

(A) -8

(B) -6

(C) 6

(D) 8